Mahdi Alam

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PROFILE

MSc Computer Science researcher with Python, PyTorch and scikit-learn experience in self-supervised learning, adaptive optimization and quantitative AI evaluation. Designs reproducible experiments and ablation studies; implements NLP retrieval baselines and an AI-assisted local document-search API prototype.

EDUCATION

Toronto Metropolitan University | Toronto, ON

MSc, Computer Science | Thesis research in artificial intelligence
September 2023 - Expected December 2026 (possible extension to January 2027)

BSc (Honours), Computer Science | September 2019 - May 2023

TECHNICAL SKILLS

Programming: Python, C++, C

Data and machine learning: Pandas, NumPy, SciPy, Scikit-learn, PyTorch, TensorFlow, OpenCV

Methods: Experiment design, quantitative evaluation, ablation studies, clustering, self-supervised learning, optimization

Tools: Git, Docker, Linux

RESEARCH EXPERIENCE

MSc Thesis Research - Self-Supervised Medical Image Registration
Toronto Metropolitan University | Toronto, ON

- Designed a two-phase, coarse-to-fine training strategy combining multi-resolution curriculum learning and adaptive optimization for self-supervised deep learning.
- Improved Dice Similarity Coefficient by up to 1.7 points across public medical-image datasets (LPBA40, IXI, OASIS), using quantitative benchmark evaluation.
- Evaluated deformation quality through Jacobian determinant analysis and conducted reproducible Python experiments with ablation studies.
- Extended On-the-Fly Guidance without changing model architecture or inference pathways. Research manuscript in preparation, not published.

SELECTED PROJECTS

Document Search API - AI-assisted learning prototype | Python | October 2026

- Built BM25 document ranking and JSON HTTP endpoints for search, health checks, and extractive evidence responses with source identifiers.
- Implemented query validation and request limits; repository includes HTTP integration tests. Local demonstration prototype.

Search Engine - NLP and Retrieval | Python | October-November 2022

- Implemented TF-IDF and BM25 retrieval; trained Naive Bayes and linear SVM baselines and compared retrieval effectiveness.

Model Lab | Python, JavaScript, GitHub Actions | October 2026

New AI-assisted learning project | mahdialam.com/model-lab/

- Trained a deterministic logistic classifier with train-only scaling, exported model parameters and provenance, and built a validated serverless inference API. Added reproducibility and API checks: 15 local tests passed.
- Evaluated on 200 held-out synthetic samples: 93% accuracy versus 52% majority baseline. Synthetic demonstration only; not enterprise MLOps or real-world sensor validation.

AVAILABILITY

Canadian citizen; authorized to work in Canada without sponsorship. Toronto-based; open to Canadian remote work or relocation within Canada. Available one month after an offer. MSc expected December 2026, possibly January 2027.